

PSYC 51.17: Models of Language and Communication

Lecture 27: Final Project Work Session

Models of Language and Conversation

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Today's Agenda

Where We Are

The Home Stretch

This is it—the final week of the course! Your final projects are due soon, and presentations are coming up.

This week's schedule:

- **Today (Monday):** Project work session, debugging support, presentation prep
- **Wednesday:** Final project presentations (Part 1)
- **Friday:** Final project presentations (Part 2)
- **End of Friday:** All final materials due (11:59 PM Eastern)

2. **Video Presentation:** 10-12 minute video (YouTube link)

- Cover motivation, approach, results, discussion
- All team members participate

3. **Written Writeup:** 2-5 page report

Presentation Format

Video Component:

- **Length:** 10-12 minutes
- **Format:** Pre-recorded video
- **Platform:** YouTube (unlisted OK)

Content:

- Motivation (why?)

In-Class Component:

- **Show video** to class
- **5-10 min Q&A**/discussion
- **All team members** present
- Be ready for questions about:
 - Technical details

What Makes a Great Presentation?

Content:

- **Clear motivation:** Why does this problem matter?
- **Technical depth:** Show you understand the methods
- **Honest results:** Good and bad—what worked, what didn't?
- **Critical analysis:** Limitations, future work, broader implications

Presentation quality:

- **Clear visuals:** Readable slides, effective figures

2. Motivation & Background (2 min)

- The problem, why it matters, research questions

3. Approach & Methods (3-4 min)

- Models, datasets, techniques, implementation details

4. Results (3-4 min)

(no errors!)

- ☐ All required packages installed with `!pip install` commands
- ☐ Dataset downloads automated (no manual file uploads)
- ☐ Random seeds set for reproducibility
- ☐ Code is well-commented and organized
- ☐ Markdown cells explain what's happening and why

Results & Analysis:

- ☐ All experiments completed and documented

Testing Reproducibility

Critical Step: Test in Clean Environment

Before submitting, open a **brand new** Colab session and run your notebook from top to bottom. This is what graders will do!

Common reproducibility issues:

1. **Missing installations:** Did you forget a `!pip install` ?
2. **File paths:** Are you referencing files that only exist on your machine?
3. **API keys:** Did you hard-code keys? (Use environment variables or instructions)

Code Quality Matters

Good code is:

- **Readable:** Clear variable names, logical organization
- **Documented:** Comments, docstrings, markdown explanations
- **Modular:** Functions for repeated tasks, not copy-paste code
- **Efficient:** Reasonable runtime, not wasteful

Example of good documentation:

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```
1 | def compute_embeddings(texts, model_name="sentence-transformers/all-
```

- **Cherry-picking:** Only showing best results, hiding failures
- **Ignoring errors:** Sweeping bugs under the rug instead of fixing
- **Last-minute testing:** Discovering critical bugs right before deadline

Presentation pitfalls:

- **Too much detail:** Getting lost in implementation minutiae
- **No motivation:** Jumping straight to methods without

When Things Don't Work

Research Reality

Not everything will work perfectly. That's **completely normal** in research!

If experiments fail or results are negative:

- **Don't panic!** Negative results are still valuable
- **Analyze why:** What went wrong? What did you learn?
- **Be honest:** Discuss failures and limitations openly
- **Show process:** Document what you tried and why
- **Suggest improvements:** What would you do differently?

2. **Priority 2 (High):** Create presentation video

- Draft slides
- Record and edit video
- Upload to YouTube

3. **Priority 3 (Medium):** Polish notebook

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Grading Breakdown

How your project will be evaluated:

Results \	Analysis	20%
Presentation (Video + In-Class)	15%	
Written Writeup	10%	
Proposal	10%	

What We're Looking For

Excellent projects demonstrate:

- **Technical sophistication:** Mastery of course concepts
- **Rigor:** Well-designed experiments, appropriate evaluation
- **Critical thinking:** Thoughtful analysis of results and limitations
- **Clarity:** Clear communication in code, writing, and presentation
- **Reproducibility:** Everything runs and can be verified
- **Insight:** Novel findings or meaningful contributions

Think about it!

Would you be proud to show this project in a job interview or grad school

Today's Work Session

Use today for:

- **Debugging:** Fix critical bugs, resolve issues
- **Completing experiments:** Finish any remaining analyses
- **Creating presentation:** Draft slides, record video
- **Testing reproducibility:** Run notebook in clean Colab
- **Peer feedback:** Show work to other teams, get suggestions
- **Getting help:** Ask questions, troubleshoot with instructor

before

3. **Check documentation:** Are you using the API correctly?
4. **Simplify:** Create minimal example that reproduces the issue
5. **Use GenAI tools:** Claude, ChatGPT can help debug
6. **Ask for help:** Discord, peers, instructor

If you're stuck on project direction:

1. **Review your proposal:** What was your original plan?

Key Takeaways

1. **Prioritize ruthlessly:** Focus on what matters most
2. **Test reproducibility:** Run in clean Colab before submitting
3. **Be honest about limitations:** Negative results are OK
4. **Practice your presentation:** Time it, refine it, polish it
5. **Use available support:** Office hours, peers, Discord
6. **Start with working code:** Polish can come later
7. **Document as you go:** Don't leave it all for the last minute

Remember

This is the culmination of everything you've learned. From ELIZA to GPT, from

Before you submit Friday night, verify:

- ☐ Code runs completely in fresh Colab (tested!)
- ☐ Video uploaded to YouTube, link included
- ☐ Writeup is 2-5 pages with all required sections
- ☐ All team members listed on all deliverables
- ☐ References properly cited
- ☐ Presentation ready for Wednesday or Friday
- ☐ All files submitted to Canvas

Submit early if possible! Don't wait until 11:59 PM—
technical issues happen.

You've Got This!

You've learned an incredible amount this quarter.

You started with simple string manipulation and
pattern matching...

And now you're building sophisticated LLM systems!

Trust your skills. Use your resources. Do great work.

We can't wait to see your presentations!

****Now: Let's get to work! ****